

# Introduction to Data Science

## Group assignment 3

Uppsala University

Michał Kamiński  
Stanisław Kamiński  
Gaston Törnquist  
Marwan Tawfeek  
Radosław Gersz

# 1 Exercise 1

a) Transition matrix

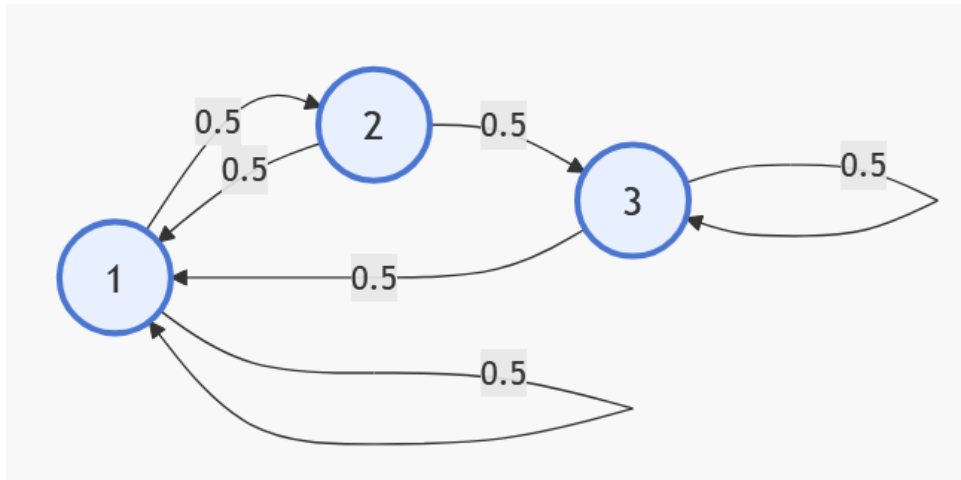


Figure 1: Transition matrix from task 1 done in Mermaid

b) Stationary distribution

To find the stationary distribution  $\pi = [\pi_1, \pi_2, \pi_3]$  analytically, we must solve the system of linear equations defined by  $\pi P = \pi$  subject to the constraint  $\sum \pi_i = 1$ . Based on the transition matrix  $P$ , the equations are:

$$\pi_1 = 0.5\pi_1 + 0.5\pi_2 + 0.5\pi_3$$

$$\pi_2 = 0.5\pi_1$$

$$\pi_3 = 0.5\pi_2 + 0.5\pi_3$$

However, we can realize that the stationary distribution  $\pi$  is the left eigenvector (corresponding to eigenvalue 1) of our transition matrix. Thus, it can be found with the following Python code:

```

import numpy as np
# define the transition matrix
P = np.array([[0.5, 0.5, 0], [0.5, 0, 0.5], [0.5, 0, 0.5]])

eigenvals, eigenvecs = np.linalg.eig(P.T)

# find the index of the eigenvalue that is 1
index = np.isclose(eigenvals, 1)

stat = eigenvecs[:, index].flatten()
stat = stat / np.sum(stat)

```

The stationary distribution  $\pi$  turns out to be:

$$\pi = [0.5, 0.25, 0.25]$$

- c) Probability of reaching state 2 from state 1 at time 4

Since we start at time  $t=1$ , we need to make steps to reach time 4. This is equivalent to calculating:

$$P_4 = P^3$$

Doing this in Python we obtain:

$$P_4 = \begin{pmatrix} 0.5 & 0.25 & 0.25 \\ 0.5 & 0.25 & 0.25 \\ 0.5 & 0.25 & 0.25 \end{pmatrix}$$

The probability of reaching state 2 from 1 at time 4 is in the top row and in the middle column with a value 0.25.

- d) Expected hitting time of state 3

Let  $k_i$  denote the expected number of steps to reach state 3 for the first time, starting from state  $i$ . We want to find  $k_1$ . Since state 3 is the target,  $k_3 = 0$ . For the other states, we use the first-step analysis equation  $k_i = 1 + \sum_j P_{ij} k_j$ .

The system of equations is:

$$k_1 = 1 + 0.5k_1 + 0.5k_2$$

$$k_2 = 1 + 0.5k_1 + 0.5k_3$$

Substituting  $k_3 = 0$  into the second equation gives  $k_2 = 1 + 0.5k_1$ . We can now substitute this expression for  $k_2$  back into the first equation:

$$k_1 = 1 + 0.5k_1 + 0.5(1 + 0.5k_1)$$

$$k_1 = 1 + 0.5k_1 + 0.5 + 0.25k_1$$

$$k_1 = 1.5 + 0.75k_1$$

Solving for  $k_1$ :

$$0.25k_1 = 1.5$$

$$k_1 = 6$$

The expected time until the chain reaches state 3 given it starts in state 1 is 6 steps.

- e) Period of each state

The period of a state  $i$  is defined as the greatest common divisor (GCD) of the number of steps in all possible return paths to  $i$ . Looking at the transition matrix, we observe that states 1 and 3 have self-loops (i.e.,  $P_{11} > 0$  and  $P_{33} > 0$ ).

- Since state 1 can return to itself in 1 step,  $d(1) = \text{GCD}(1, \dots) = 1$ .
- Similarly, state 3 has a period of 1.

Since the Markov chain is irreducible (it is possible to reach any state from any other state), all states in the chain must have the same period. Therefore, state 2 also has a period of 1.

**Conclusion:** The period of each state is 1. The chain is aperiodic.

## 2 Exercise 2

### a) Empirical versions of precision and recall

The **empirical precision** estimates the probability that a positive prediction is correct:

$$\widehat{\text{Precision}} = \frac{\sum_{i=1}^n \mathbb{1}(Y_i = 1, g(X_i) = 1)}{\sum_{i=1}^n \mathbb{1}(g(X_i) = 1)}$$

This is the ratio of true positives to all positive predictions.

The **empirical recall** estimates the probability that an actual positive case is correctly identified:

$$\widehat{\text{Recall}} = \frac{\sum_{i=1}^n \mathbb{1}(Y_i = 1, g(X_i) = 1)}{\sum_{i=1}^n \mathbb{1}(Y_i = 1)}$$

This is the ratio of true positives to all actual positive cases.

### b) Expected cost formula

In the battery health monitoring context,  $Y = 1$  means the battery has deteriorated,  $Y = 0$  means it is healthy,  $g(X) = 1$  means we predict deteriorated (run test with cost  $c$ ), and  $g(X) = 0$  means we predict healthy (no test). If the battery is deteriorated but not tested, it dies during use with cost  $d$ .

Define the cost random variable  $C$  associated with the decision  $g(X)$ :

$$C = \begin{cases} c & \text{if } g(X) = 1 \\ d & \text{if } g(X) = 0 \text{ and } Y = 1 \\ 0 & \text{if } g(X) = 0 \text{ and } Y = 0 \end{cases}$$

The expected cost can be computed by considering all possible outcomes:

$$\begin{aligned} \mathbb{E}[C] &= c \cdot \mathbb{P}(g(X) = 1) + d \cdot \mathbb{P}(g(X) = 0, Y = 1) \\ &= c \cdot \mathbb{P}(g(X) = 1) + d \cdot \mathbb{P}(Y = 1) \cdot \mathbb{P}(g(X) = 0 \mid Y = 1) \end{aligned}$$

Since  $\mathbb{P}(g(X) = 0 \mid Y = 1) = 1 - \mathbb{P}(g(X) = 1 \mid Y = 1) = 1 - \text{Recall}$ , we have:

$$\mathbb{E}[C] = c \cdot \mathbb{P}(g(X) = 1) + d \cdot \mathbb{P}(Y = 1) \cdot (1 - \text{Recall})$$

Alternatively, we can decompose  $\mathbb{P}(g(X) = 1)$  by conditioning on  $Y$ :

$$\begin{aligned} \mathbb{E}[C] &= c \cdot [\mathbb{P}(g(X) = 1 \mid Y = 1) \cdot \mathbb{P}(Y = 1) \\ &\quad + \mathbb{P}(g(X) = 1 \mid Y = 0) \cdot \mathbb{P}(Y = 0)] \\ &\quad + d \cdot (1 - \text{Recall}) \cdot \mathbb{P}(Y = 1) \end{aligned}$$

This can be written as:

$$\mathbb{E}[C] = c \cdot \text{Recall} \cdot \mathbb{P}(Y = 1) + c \cdot \text{FPR} \cdot \mathbb{P}(Y = 0) + d \cdot (1 - \text{Recall}) \cdot \mathbb{P}(Y = 1)$$

where  $\text{FPR} = \mathbb{P}(g(X) = 1 \mid Y = 0)$  is the false positive rate.

Simplifying further:

$$\mathbb{E}[C] = \mathbb{P}(Y = 1) \cdot [c \cdot \text{Recall} + d \cdot (1 - \text{Recall})] + c \cdot \text{FPR} \cdot \mathbb{P}(Y = 0)$$

c) Confidence intervals

### Confidence Interval for Expected Cost

The empirical expected cost is:

$$\widehat{\mathbb{E}[C]} = \frac{1}{n} \sum_{i=1}^n C_i = \frac{c}{n} \sum_{i=1}^n \mathbb{1}(g(X_i) = 1) + \frac{d}{n} \sum_{i=1}^n \mathbb{1}(g(X_i) = 0, Y_i = 1)$$

where  $C_i$  is the cost associated with the  $i$ -th test observation.

By the Central Limit Theorem, for large  $n$ :

$$\frac{\sqrt{n}(\widehat{\mathbb{E}[C]} - \mathbb{E}[C])}{\widehat{\sigma}} \xrightarrow{d} \mathcal{N}(0, 1)$$

where  $\widehat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (C_i - \widehat{\mathbb{E}[C]})^2$  is the sample variance of the costs.

An approximate  $(1 - \alpha)$  confidence interval for the expected cost is:

$$\widehat{\mathbb{E}[C]} \pm z_{\alpha/2} \cdot \frac{\widehat{\sigma}}{\sqrt{n}}$$

where  $z_{\alpha/2}$  is the  $(1 - \alpha/2)$  quantile of the standard normal distribution (e.g.,  $z_{0.025} = 1.96$  for a 95% confidence interval).

### Confidence Interval for Precision

Let  $n_1 = \sum_{i=1}^n \mathbb{1}(g(X_i) = 1)$  be the number of positive predictions. The empirical precision can be viewed as a sample proportion from a binomial distribution.

Using the normal approximation for large  $n_1$ , an approximate  $(1 - \alpha)$  confidence interval for precision is:

$$\widehat{\text{Precision}} \pm z_{\alpha/2} \sqrt{\frac{\widehat{\text{Precision}}(1 - \widehat{\text{Precision}})}{n_1}}$$

### Confidence Interval for Recall

Let  $n_+ = \sum_{i=1}^n \mathbb{1}(Y_i = 1)$  be the number of actual positive cases in the test set. Similarly, using the normal approximation, an approximate  $(1 - \alpha)$  confidence interval for recall is:

$$\widehat{\text{Recall}} \pm z_{\alpha/2} \sqrt{\frac{\widehat{\text{Recall}}(1 - \widehat{\text{Recall}})}{n_+}}$$

### Validity of Normal Approximations

These confidence intervals based on normal approximations are valid when:

- For expected cost:  $n$  is sufficiently large (typically  $n \geq 30$ )
- For precision:  $n_1$  is sufficiently large and  $n_1 \cdot \widehat{\text{Precision}} \geq 5$  and  $n_1 \cdot (1 - \widehat{\text{Precision}}) \geq 5$
- For recall:  $n_+$  is sufficiently large and  $n_+ \cdot \widehat{\text{Recall}} \geq 5$  and  $n_+ \cdot (1 - \widehat{\text{Recall}}) \geq 5$

For smaller sample sizes or extreme proportions (close to 0 or 1), alternative methods such as Wilson score intervals, exact binomial confidence intervals (Clopper-Pearson), or bootstrap confidence intervals should be preferred as they provide better coverage probabilities.

### 3 Exercise 3

Let

$$X = (X_1, \dots, X_d), \quad Y = (Y_1, \dots, Y_d),$$

where

$$X_i, Y_i \sim \mathcal{N}(0, 1),$$

and all coordinates are independent. Clearly,

$$\mathbb{E}[X_i] = \mathbb{E}[Y_i] = 0, \quad \text{Var}(X_i) = \text{Var}(Y_i) = 1.$$

Recall that  $X$  and  $Y$  are orthogonal if and only if their dot product is zero:

$$X \cdot Y = \sum_{i=1}^d X_i Y_i = 0.$$

Define a new random variable

$$Z_i = X_i Y_i.$$

Since  $X_i$  and  $Y_i$  are independent,

$$\mathbb{E}[Z_i] = \mathbb{E}[X_i Y_i] = \mathbb{E}[X_i] \mathbb{E}[Y_i] = 0.$$

Next, consider the norms of the vectors. We have

$$\|X\|^2 = \sum_{i=1}^d X_i^2.$$

Since  $\mathbb{E}[X_i^2] = 1$ , by the Law of Large Numbers,

$$\frac{1}{d} \|X\|^2 = \frac{1}{d} \sum_{i=1}^d X_i^2 \xrightarrow{P} 1,$$

and therefore

$$\frac{\|X\|}{\sqrt{d}} \xrightarrow{P} 1.$$

The same argument applies to  $Y$ :

$$\frac{\|Y\|}{\sqrt{d}} \xrightarrow{P} 1.$$

From this we conclude that  $\|X\|$  and  $\|Y\|$  converge to  $\sqrt{d}$ . Now consider the empirical mean of the products:

$$\mu_d = \frac{1}{d} \sum_{i=1}^d X_i Y_i.$$

By the Law of Large Numbers,

$$\mu_d \xrightarrow{P} \mathbb{E}[Z_i] = 0,$$

which implies

$$\frac{1}{d} \sum_{i=1}^d X_i Y_i \xrightarrow{P} 0.$$

Finally, the cosine similarity between  $X$  and  $Y$  is

$$\cos(X, Y) = \frac{X \cdot Y}{\|X\| \|Y\|} = \frac{\sum_{i=1}^d X_i Y_i}{\|X\| \|Y\|}.$$

Using the previous results,

$$\cos(X, Y) = \frac{1}{d} \sum_{i=1}^d X_i Y_i \xrightarrow{P} 0.$$

Since the cosine similarity converges to zero, the vectors  $X$  and  $Y$  are asymptotically orthogonal. We can also prove this by bounding the probability that their dot product deviates from zero using Chebyshev's inequality.

By Chebyshev's inequality,

$$\mathbb{P} \left( \left| \frac{X \cdot Y}{d} \right| > \varepsilon \right) \leq \frac{\text{Var} \left( \frac{X \cdot Y}{d} \right)}{\varepsilon^2}.$$

Using the scaling property of variance,

$$\text{Var} \left( \frac{X \cdot Y}{d} \right) = \frac{1}{d^2} \text{Var}(X \cdot Y).$$

Next, compute  $\text{Var}(X \cdot Y)$ . Since the variables  $X_i Y_i$  are independent and centered,

$$\text{Var}(X \cdot Y) = \sum_{i=1}^d \text{Var}(X_i Y_i).$$

For each term,

$$\text{Var}(X_i Y_i) = \mathbb{E}[X_i^2 Y_i^2] - (\mathbb{E}[X_i Y_i])^2.$$

Because  $X_i$  and  $Y_i$  are independent standard Gaussian random variables,

$$\mathbb{E}[X_i Y_i] = 0, \quad \mathbb{E}[X_i^2 Y_i^2] = \mathbb{E}[X_i^2] \mathbb{E}[Y_i^2] = 1 \cdot 1 = 1.$$

Hence,

$$\text{Var}(X_i Y_i) = 1, \quad \text{Var}(X \cdot Y) = \sum_{i=1}^d 1 = d.$$

Therefore,

$$\text{Var} \left( \frac{X \cdot Y}{d} \right) = \frac{d}{d^2} = \frac{1}{d}.$$

Substituting into Chebyshev's inequality gives

$$\mathbb{P} \left( \left| \frac{X \cdot Y}{d} \right| > \varepsilon \right) \leq \frac{1}{d\varepsilon^2}.$$

Equivalently,

$$\mathbb{P}(|X \cdot Y| > \varepsilon d) \leq \frac{1}{d\varepsilon^2}.$$

Because the probability vanishes as  $d \rightarrow \infty$ , confirming that  $X$  and  $Y$  are almost orthogonal with high probability.

## 4 Exercise 4

(a) Fix  $i$  and let  $u_i = (u_{i1}, u_{i2}, \dots, u_{in})^\top \in \mathbf{R}^n$  be a unit vector (so  $u_i \neq 0$ ). Consider

$$A := u_i u_i^\top \in \mathbf{R}^{n \times n}.$$

Matrix  $A$  is so-called outer product, which by definition has  $\text{rank} = 1$ , but we can prove that explicitly.

When considering the way the vectors in  $A$  are multiplied, we can see that every column in  $A$  is a scalar multiple of the same vector, so the  $j$ -th column can be written as

$$A^{(j)} = u_{ij} u_i$$

, where  $u_{ij}$  is a scalar. Therefore, we can write that

$$A^{(1)} = u_{i1} u_i, \quad A^{(2)} = u_{i2} u_i, \quad \dots, \quad A^{(n)} = u_{in} u_i.$$

Therefore every column lies in  $\text{span}\{u_i\}$ , meaning the column space is at most one-dimensional. Hence

$$\text{rank}(A) \leq 1.$$

Now we only have to show that  $\text{rank}(A) \neq 0$ . Since  $u_i$  is a unit vector, at least one component  $u_j \neq 0$ . For such a  $j$ , the  $j$ -th column is

$$A^{(j)} = u_j u_i \neq 0,$$

because  $u_j \neq 0$  and  $u_i \neq 0$ . Thus  $A \neq 0$ , so  $\text{rank}(A) \neq 0$ .

Combining  $\text{rank}(A) \leq 1$  with  $\text{rank}(A) \neq 0$ , we get

$$\text{rank}(u_i u_i^\top) = 1.$$

**Range of  $A$ .** Since  $u_i u_i^\top$  is an outer product, each column of  $u_i u_i^\top$  is a scalar multiple of the vector  $u_i$ . Hence the column space is one-dimensional and is spanned by  $u_i$ , which implies

$$\text{range}(u_i u_i^\top) = \text{span}\{u_i\}.$$

**Null space of  $A$ .** Let  $x \in \mathbf{R}^n$ . Then

$$Ax = u_i u_i^\top x = u_i (u_i^\top x).$$

Thus,

$$Ax = 0 \iff u_i (u_i^\top x) = 0.$$

Since  $u_i \neq 0$ , this holds if and only if

$$u_i^\top x = 0.$$

Therefore,

$$\text{null}(u_i u_i^\top) = \{x \in \mathbf{R}^n \mid u_i^\top x = 0\}.$$

This set consists of all vectors orthogonal to  $u_i$ , so equivalently,

$$\text{null}(u_i u_i^\top) = \text{span}\{u_i\}^\perp.$$

(b) Define the matrix

$$(u_1 \ \cdots \ u_r) \in \mathbf{R}^{n \times r}.$$

Then

$$AA^\top = (u_1 \ \cdots \ u_r) \begin{pmatrix} u_1^\top \\ \vdots \\ u_r^\top \end{pmatrix} = \sum_{i=1}^r u_i u_i^\top = U$$

Since the vectors  $u_1, \dots, u_r$  are linearly independent, the columns of  $A$  are linearly independent, and therefore

$$\text{rank}(A) = r.$$

Next, we determine the null space of  $U$ . For any  $x \in \mathbf{R}^n$ ,

$$Ux = 0 \iff AA^\top x = 0 \iff x^\top AA^\top x = 0 \iff \|A^\top x\|^2 = 0 \iff A^\top x = 0.$$

Thus,

$$\text{null}(U) = \text{null}(A^\top).$$

By the rank-nullity theorem,

$$\dim(\text{null}(U)) + \text{rank}(U) = n$$

We can transform that in the following way

$$\text{rank}(U) = n - \dim(\text{null}(U)) = n - \dim(\text{null}(A^\top)) = \text{rank}(A^\top) = \text{rank}(A) = r.$$

Therefore, the matrix  $U = \sum_{i=1}^r u_i u_i^\top$  has rank  $r$ .

(c) Recall that

$$U = \sum_{i=1}^r u_i u_i^\top.$$

The matrix  $U$  is symmetric and positive semidefinite. Hence its right singular vectors coincide with its eigenvectors.

(i) **Are  $u_1, \dots, u_r$  right singular vectors of  $U$ ?**

If the right singular vectors of  $U$  coincide with its eigenvectors, we have to check if for some  $j \in \{1, \dots, r\}$  the following is true.

$$Uu_j = \lambda u_j \quad \text{for some } \lambda \in \mathbf{R}$$

We can verify that by checking what happens when we multiply this matrix with  $u_j$

$$Uu_j = \sum_{i=1}^r u_i u_i^\top u_j = \sum_{i=1}^r (u_i^\top u_j) u_i.$$

We can see that in general, this is a linear combination of all vectors  $u_1, \dots, u_r$ , and not a scalar multiple of  $u_j$ . Therefore, we can see that the vector  $u_j$  is not an eigenvector of  $U$ . Hence, in general, the vectors  $u_1, \dots, u_r$  are *not* right singular vectors of  $U$ .

(ii) **Singular values when  $u_1, \dots, u_r$  are orthogonal.**

Now we assume that the vectors  $u_1, \dots, u_r$  are orthogonal, i.e.

$$u_i^\top u_j = 0 \quad \text{for } i \neq j.$$

We now evaluate each term in the sum. For  $i \neq j$ , orthogonality implies

$$u_i u_i^\top u_j = u_i (u_i^\top u_j) = u_i \cdot 0 = 0.$$

For  $i = j$ , we have

$$u_j u_j^\top u_j = u_j (u_j^\top u_j) = \|u_j\|^2 u_j.$$

Therefore, all terms in the sum vanish except the one with  $i = j$ , and we obtain

$$U u_j = \|u_j\|^2 u_j.$$

Thus,  $u_j$  is an eigenvector of  $U$  with eigenvalue  $\|u_j\|^2$ .

If  $x$  is orthogonal to  $\text{span}\{u_1, \dots, u_r\}$ , then  $u_i^\top x = 0$  for all  $i$ , and therefore

$$Ux = 0.$$

Hence, 0 is an eigenvalue of  $U$  with multiplicity  $n - r$ .

Since  $U$  is symmetric and positive semidefinite, its singular values are equal to its eigenvalues. Therefore,

$$\sigma_1 = \dots = \sigma_r = 1, \quad \sigma_{r+1} = \dots = \sigma_n = 0.$$

## 5 Exercise 5

Let  $X \sim \text{Uniform}(B_1)$  and define  $Y = \|X\|_2$  (the Euclidean norm).

(a) Find the distribution of  $Y$ .

We let  $X$  be defined in some space  $\mathbb{R}^n$ . We define the distribution function (CDF):

$$F_Y(r) = P(Y \leq r) \tag{1}$$

$$= P(\|X\|_2 \leq r) \tag{2}$$

$$= P(X \in B_r) \tag{3}$$

Intuitively, this can be thought of as the probability of  $Y$  falling within some sub-region  $B_r \subseteq B_1$  with radius  $r \in [0, 1]$ . We know from **Model 9.11** (of our lecture notes) that the probability of  $Y$  falling into a given 'sub-ball'  $B_r$  is given by:

$$F_Y(r) = \frac{\text{Volume}(B_r)}{\text{Volume}(B_1)}$$

Using the formula  $\frac{\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2}+1)} r^n$  for the volume of an  $n$ -dimensional ball of radius  $r$ , we find that:

$$F_Y(r) = \frac{\text{Volume}(B_r)}{\text{Volume}(B_1)} = \frac{r^n}{1^n} = r^n$$

Thus, the distribution of  $Y$  is  $F_Y(y) = y^n$ , where  $n$  is the dimension of the space  $\mathbb{R}^n$ .

(b) What is the distribution of  $\ln(1/Y)$ ?

Let  $Z = \ln(1/Y)$ . We wish to find the distribution of  $Z$ , that is:

$$F_Z(z) = P(Z \leq z) \quad (4)$$

$$= P(\ln(\frac{1}{Y}) \leq z) \quad (5)$$

$$= P(\frac{1}{Y} \leq e^z) \quad (6)$$

$$= P(Y \geq e^{-z}) \quad (7)$$

for  $Y \in (0, 1]$  and  $z \geq 0$ .

We know from part (a) that  $P(Y \leq y) = y^n$ . Applying the complement rule, and using the fact that  $P(Y \leq a) = P(Y < a)$  for continuously defined  $Y$ , we get that:

$$P(Y \geq e^{-z}) = 1 - P(Y < e^{-z}) \quad (8)$$

$$= 1 - P(Y \leq e^{-z}) \quad (9)$$

$$= 1 - e^{-zn} \quad (10)$$

Thus, the distribution of  $\ln(1/Y)$  is given by:

$$F_Z(z) = 1 - e^{-zn}$$

where  $Z = \ln(1/Y)$ ,  $n$  is the dimension of our space  $\mathbb{R}^n$  and  $z \geq 0$ .

(c) Calculate  $\mathbb{E}[\ln(1/Y)]$ , (i) first by using the distribution function of  $Y$  and (ii) then by using the distribution function of  $\ln(1/Y)$ .

(i) To find  $\mathbb{E}[\ln(1/Y)]$  using the distribution function of  $Y$ , we utilize **Definition 2.53**, for the expectation of a function  $g(X)$  on a continuous R.V.  $X$  with PDF  $f(x)$ :

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

Let  $g(Y) = Z = \ln(1/Y)$ . To apply the formula above, we first derive the PDF  $f_Y(y)$  as:

$$f_Y(y) = F'_Y(y) = \frac{d}{dy} y^n = ny^{n-1}$$

Next, we apply the formula from **Definition 2.53**:

$$\mathbb{E}[\ln(1/Y)] = \mathbb{E}[g(Y)] = \int_{-\infty}^{\infty} g(y) f_Y(y) dy \quad (11)$$

$$= \int_0^1 \ln(\frac{1}{y}) ny^{n-1} dy \quad (12)$$

$$= n \int_0^1 \ln(\frac{1}{y}) y^{n-1} dy \quad (13)$$

where we set our integration limits to reflect the domain  $Y \in (0, 1]$ . Let  $I = \int \ln(\frac{1}{y}) y^{n-1} dy$ , such that  $\mathbb{E}[\ln(1/Y)] = n [I]_0^1$ .

Next, let  $u = \ln(1/y)$  and let  $dv = y^{n-1} dy$  such that  $I = \int u dv$ .

We find that:

$$du = -y^{-2} (\frac{1}{y})^{-1} dy = -\frac{1}{y} dy$$

and

$$v = \int dv = \int y^{n-1} dy = \frac{y^n}{n}$$

where we ignore integration constants as we will later evaluate the definite integral  $[I]_0^1$ . Applying integration by parts, we get that:

$$I = uv - \int v du = \ln(\frac{1}{y}) \frac{y^n}{n} - \int (\frac{y^n}{n}) (\frac{-1}{y}) dy \quad (14)$$

$$= \ln(\frac{1}{y}) \frac{y^n}{n} + \frac{1}{n} \int y^{n-1} dy \quad (15)$$

$$= \ln(\frac{1}{y}) \frac{y^n}{n} + \frac{y^n}{n^2} + C \quad (16)$$

Finally, we evaluate the definite integral  $n [I]_0^1$ :

$$\mathbb{E}[\ln(1/Y)] = n [I]_0^1 \quad (17)$$

$$= \left[ y^n \ln(\frac{1}{y}) + \frac{y^n}{n} \right]_0^1 \quad (18)$$

$$= 0 + \frac{1}{n} + \lim_{y \rightarrow 0} y^n \ln(\frac{1}{y}) + 0 \quad (19)$$

$$= \frac{1}{n} + \lim_{y \rightarrow 0} y^n \ln(\frac{1}{y}) \quad (20)$$

Letting  $h(y) = \ln(\frac{1}{y})$ ,  $k(y) = y^{-n}$ , and noting that both functions approach infinity as  $y \rightarrow 0$  (for  $n \geq 1$ ), we apply L'Hopital's Rule to solve the expression:

$$\lim_{y \rightarrow 0} y^n \ln\left(\frac{1}{y}\right) = \lim_{y \rightarrow 0} \frac{h(y)}{k(y)} \quad (21)$$

$$= \lim_{y \rightarrow 0} \frac{h'(y)}{k'(y)} \quad (22)$$

$$= \lim_{y \rightarrow 0} \frac{-(y^{-2})\left(\frac{1}{y}\right)^{-1}}{-ny^{-n-1}} \quad (23)$$

$$= \lim_{y \rightarrow 0} \frac{y^{-1}}{ny^{-n-1}} \quad (24)$$

$$= \lim_{y \rightarrow 0} \frac{y^n}{n} = 0 \quad (25)$$

Plugging this back into Eq. (20), we find that

$$\mathbb{E}[\ln(1/Y)] = \frac{1}{n}$$

(ii) To instead find the expectation using the distribution function of  $\ln(1/Y)$ , we utilize **Definition 2.30** for the expectation of a continuous R.V.  $X$  with density  $f$ :

$$\mathbb{E}[X] = \int x f(x) dx$$

To apply this to our problem, we derive the density  $f_Z$  of  $Z = \ln(1/Y)$ :

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \frac{d}{dz} (1 - e^{-zn}) = ne^{-zn}$$

We note that  $Z \in [0, \infty)$  since  $Y \in [0, 1]$  and  $Z = \ln(1/Y)$ . This gives us the integration limits we will apply.

We now find the expectation of  $Z = \ln(1/Y)$  using **Definition 2.30**:

$$\mathbb{E}[\ln(1/Y)] = \mathbb{E}[Z] = \int_0^\infty z f_Z(z) dz = n \int_0^\infty z e^{-zn} dz$$

Let  $u_1 = z$ ,  $dv_1 = e^{-zn} dz$  and  $I_1 = u_1 dv_1$ , such that  $\mathbb{E}[\ln(1/Y)] = n [I_1]_0^\infty$ .

We have that:

$$du_1 = dz \quad \text{and} \quad v_1 = \int e^{-zn} dz = \frac{-e^{-zn}}{n}$$

Thus,

$$I_1 = u_1 v_1 - \int v_1 du_1 \quad (26)$$

$$= \frac{-ze^{-zn}}{n} - \int \frac{-e^{-zn}}{n} dz \quad (27)$$

$$= \frac{-ze^{-zn}}{n} - \frac{e^{-zn}}{n^2} \quad (+C) \quad (28)$$

$$(29)$$

Thus,

$$\mathbb{E}[\ln(1/Y)] = n [I_1]_{z=0}^{\infty} = \left[ -ze^{-zn} - \frac{e^{-zn}}{n} \right]_0^{\infty} \quad (30)$$

$$= 0 - 0 - (0 - \frac{1}{n}) \quad (31)$$

$$= \frac{1}{n} \quad (32)$$